



Unit-4 Ethics in Engineering



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Topics to discuss:

- ▶ Scope of Engineering Ethics
- ▶ Accepting and Sharing Responsibilities
- ▶ Responsible Professionals and Ethical Corporations
- ▶ Resolving Ethical Dilemmas
- ▶ Making Moral Choices

Scope of Engineering Ethics



Why Study Engineering Ethics?

- Because it is the right thing to do.
- Because it is required.
- Because it is part (a crucial part) of being a modern engineer or scientist
- Because it may save lives.
- Because it may keep you out of trouble.
- Because it may protect some tender parts of your body or soul from legal assault.

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Scope of Engineering Ethics:

- ▶ Ethics is the study of moral principles that governs a person's behaviour or conducting an activity.
- ▶ 'Engineering ethics' is the rules and standards that govern the decisions and actions of an engineer as a professional.
- ▶ During their engineering studies, the student receive the knowledge of basic science and its application, problem-solving methodology and designing, but generally receives little training in business practices, safety and ethics.
- ▶ Engineers face situations where they have to deal with the many things which are not known to them as a result of a design of new device or product.

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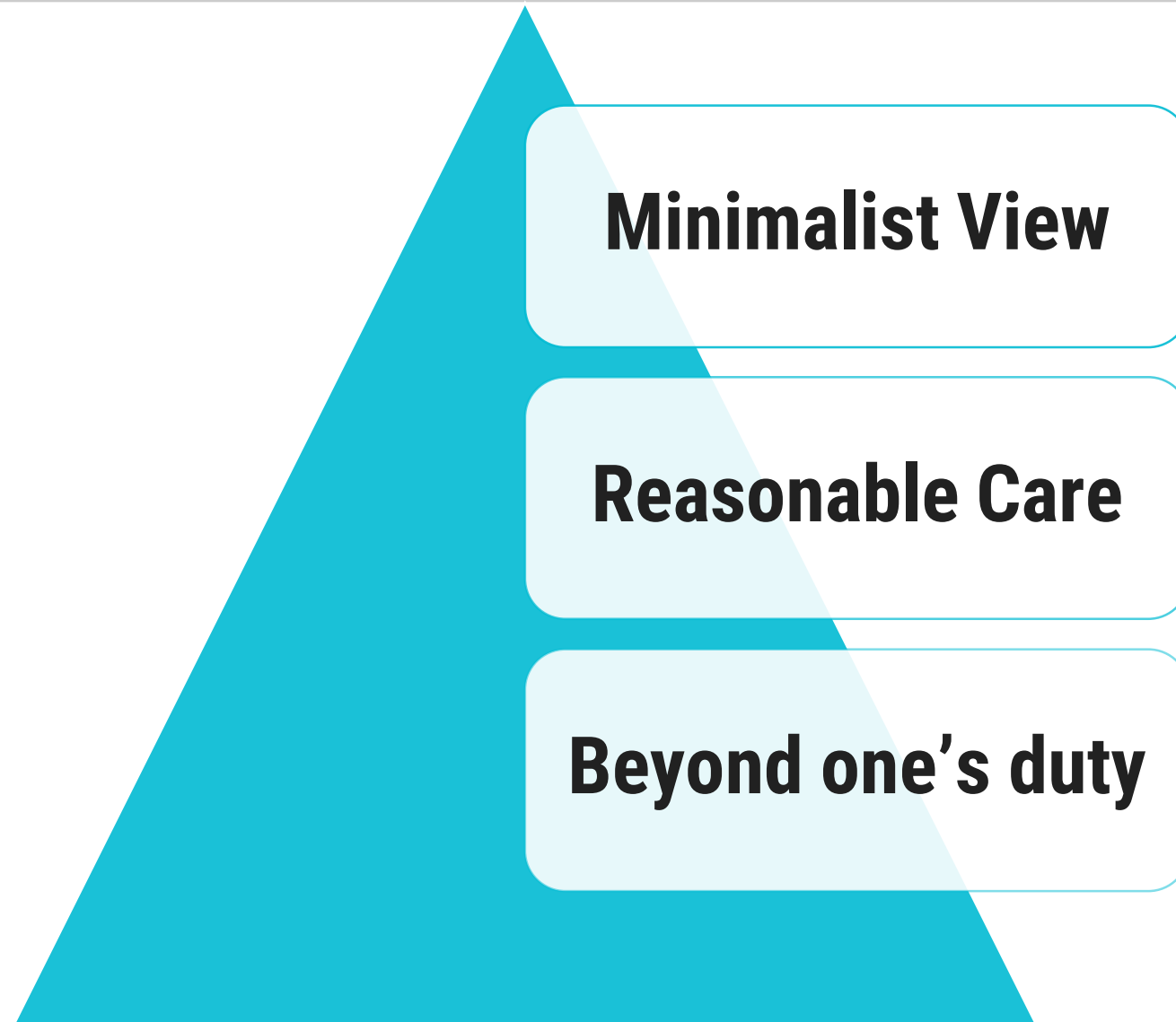
Accepting and Sharing Responsibilities



Responsibilities:

- ▶ There are various conceptions of responsibilities based on how engineers accept responsibility or what others expect them to do as a part of their responsibility.
- ▶ Moreover, when a harm is done, the responsibility is distinguished as:
 1. **Intentionally causing harm (knowingly and deliberately)**
 2. **Negligently causing harm (unknowingly but failing to exercise due care)**
 3. **Recklessly harm causing harm (having conscious awareness that harm may occur, but neglecting it without any intention of causing harm)**

How an engineer views her/his responsibility?



Whistleblowing – a right and responsibility of an engineer:

- ▶ Whistleblowing is an act by an employee for informing public or higher management of unethical or illegal behaviour by an employer or supervisor.
- ▶ There can be internal whistleblowing where an employee surpasses the immediate supervisor and reports to the higher management about the wrongdoing and external whistleblowing where the employee reports the unethical practice of one's organization to either media or law-enforcement authorities.
- ▶ Whistleblowing can be considered as the responsibility of the engineer to make others aware about the unethical practices which may harm the public.

- ▶ It is also the right, wherein the engineer can be protected for the consequences that he might face.
- ▶ But, whistleblowing must be done only when there is a dire need to do so.
- ▶ Whistleblowing must only be done, if you have already tried to put your point forward to your immediate authority and they haven't considered it seriously.

Impediments to Responsibility:

-  **Self – interest**
-  **Fear**
-  **Self-deception**
-  **Ignorance**
-  **Different Perspective**
-  **Lack of Authority**
-  **Groupthink**

Responsible Professionals and Ethical Corporations



What is Profession?

- ▶ The work that requires sophisticated skills, the use of judgement and the exercise of discretion, which is not routine and is not capable of being mechanized and significant public good results from it, it is called a **profession**.
- ▶ To be a professional, one requires extensive formal education and not simply training or apprenticeship and there are societies or organizations that are controlled by members of the profession who set the standards of conduct for professionals and admissions to professions.

Engineering as a profession:



Professional Responsibilities:



1. Confidentiality:

- ➔ In engineering, it is an engineer's obligation to keep the information of the employer and client confidential.
- ➔ Any information of an organization, like test results and data, unreleased product and designs, how the business is run, etc. can be used by competitor to gain advantage and hence must be kept confidential in interest of the organization.
- ➔ Engineers working in government departments especially for defence industry have even more stringent requirement to keep confidentiality.

2. Conflict of interest:

- ➔ The engineer may compromise the need of the public for one's own interest and this may result into various accidents occurring due to engineering/design failure.
- ➔ Taking bribes to give permissions for work that is not within the code of ethics or illegal, sharing the information of tender with a company where you have financial or other interest, or avoiding more work to go into the depth of the case are some example of giving in to self-interest at the cost of public or organizational interest.

3. Environmental Ethics:

- ➔ The detrimental effects of use of technology has made the societies and corporations more alert about environmental ethics and hence have laid down various norms for the organizations and engineers to carry out their responsibilities for environment protection.
- ➔ For example, there are standards for minimum level of certain gases and chemicals that are released by factories.

4. Computer Ethics:

- ➔ With the increasing use of computers in all fields, it provides a challenge of unethical use of it.
- ➔ Computers can be used for robbery, crime and blackmailing others.
- ➔ There is privacy issue for the information about individuals and organizations.
- ➔ Moreover, hacking is used to access private information and change or destroy some important information.

Resolving Ethical Dilemmas



Resolving Ethical Dilemmas:

- ▶ Dilemmas: Often engineers face situations while taking decisions which are morally right or wrong.
- ▶ To resolve this dilemma, following techniques are used:
 1. Line Drawing
 2. Flow Charting

1. Line Drawing:

- ▶ In this approach, a line is drawn along which various examples and hypothetical situations are placed.
- ▶ At one end, is placed 'positive paradigm' and on the other 'negative paradigm'.
- ▶ Then, all the hypothetical situations and examples are listed and placed on the line at a distance based on how near it is to positive or negative paradigm.
- ▶ Let us see it through an example.

Example

Take an example of a company that dispose a slightly hazardous waste by dumping it into a nearby lake. A residential area nearby the lake takes its drinking water supply from the lake. The average concentration of waste in the lake would be 5 parts per million (5ppm) and at this level there is no harm to the health of the consumers and they would not even be able to detect the compound in their drinking water.

For this, let's establish hypothetical examples which we should consider before taking decision:

1. Though at 5ppm it is harmless, the taste of water may change.
2. The chemical can be effectively removed by existing equipment.
3. The company will have to purchase a new equipment to remove the chemical.
4. Occasionally people may feel sick, but it would not last for more than a week.
5. Equipment can be installed at the plant to further reduce the waste level to 1 ppm.

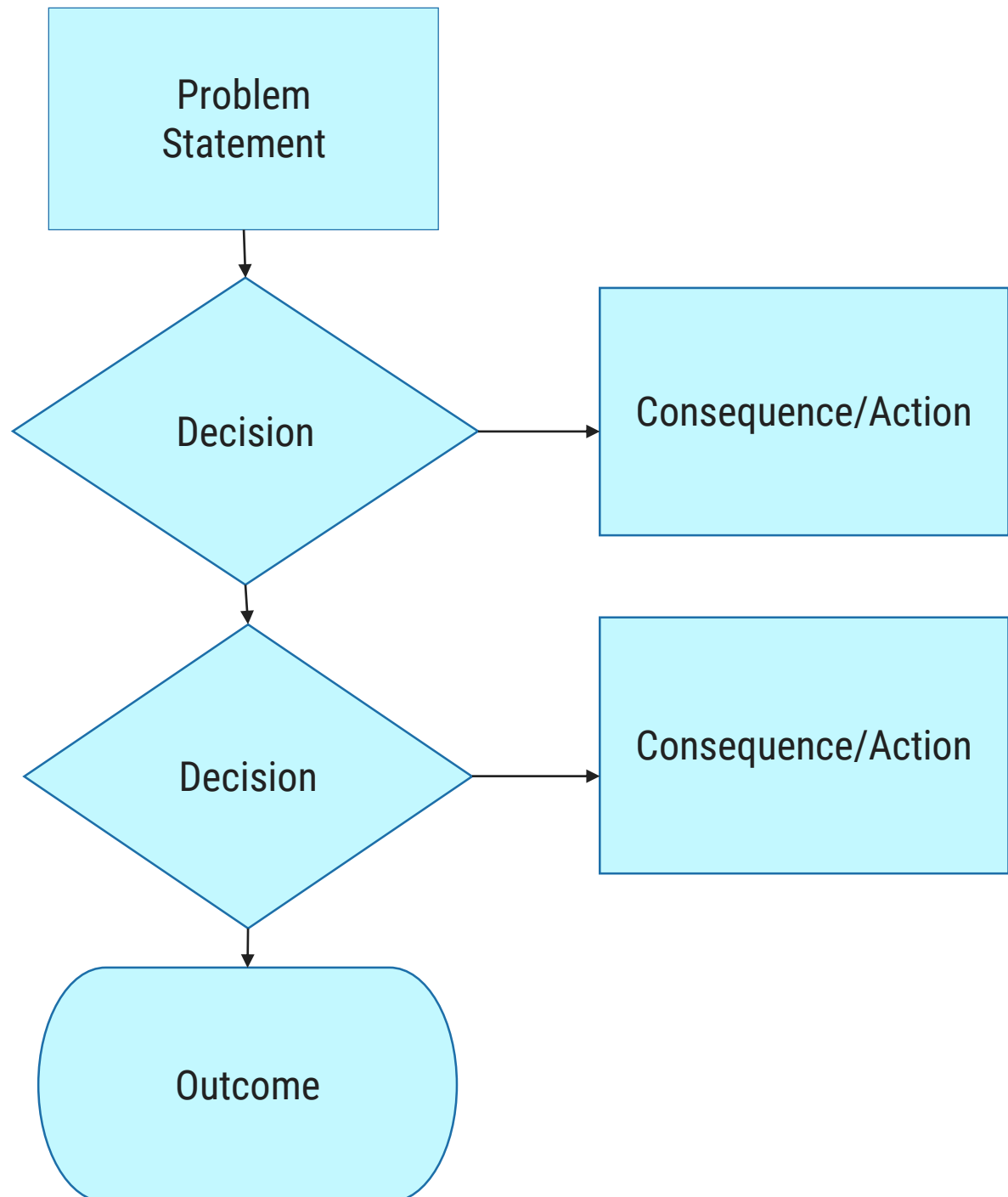


Line drawing method:

- ▶ Points towards the right contribute to ethical decisions.
- ▶ If more points are on the left hand side, then we may not go ahead with the plan, but here the decision may not be completely perfect as some situations depends on many factors which have not been taken into consideration.
- ▶ Moreover, this is not a fool-proof ethical decision making method as one may prove something to be correct when it is completely wrong.
- ▶ Hence, if not used honestly, it may fail.

2. Flow charting:

- ▶ Engineering students are used to flow charting method of programming purpose.
- ▶ Used to solve ethical dilemmas.
- ▶ Provides visual picture of a situation.
- ▶ Helps to establish sequences, identify moral issues and consequences of actions. (See in the figure which is given in next slide)
- ▶ When situation is more complicated, this chart helps to take right decision.
- ▶ Let us see how we can make these moral choices.





Making Moral Choices



Making Moral Choices:

- ▶ ‘Moral’ refers to the principles of right and wrong behavior.
- 1. The Utilitarian Approach
- 2. The right Approach
- 3. The Fairness or Justice Approach
- 4. The Common Good Approach
- 5. The Virtue approach

1. The Utilitarian Approach:

- ▶ The ethical action is the one that balance good over harm.
- ▶ Example of Terrorism
- ▶ It deals with consequences; it tries both to increase the good done and to reduce the harm done.
- ▶ If the benefit is more than the cost involved, then it can be considered to be morally correct.

2. The right Approach:

- ▶ The ethical action is the one that best protects and respects the moral rights of those affected.
- ▶ This approach starts from the belief.
- ▶ The debated list of moral rights.
- ▶ It is often said that rights imply duties – in particular, the duty to respect others' rights.

3.The Fairness or Justice Approach:

- ▶ All equals should be treated equally.
- ▶ People are paid more based on their hard work or the greater amount that they contribute to an organization
- ▶ Debate over CEO salaries

4. The Common Good Approach:

- ▶ The interlocking relationships of society are the basis of ethical reasoning and that respect and compassion for all others.
- ▶ Give attention to the common conditions that are important to the welfare of everyone.
- ▶ This may be
 - system of laws,
 - effective police and fire departments,
 - health care,
 - a public educational system,
 - public recreational areas.

5. The Virtue approach:

- ▶ A very ancient approach
- ▶ Ethical actions ought to be consistent with certain ideal virtues that provide for the full development of our humanity.
- ▶ Examples of Virtues: Honesty, courage, compassion, generosity, tolerance, love, fidelity, integrity, fairness, self-control, and prudence etc.
- ▶ “What kind of person will I become if I do this?”
or
“Is this action consistent with my acting at my best?”

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